

Active Learning and VAE on Raman Spectra to Determine Tissue Type

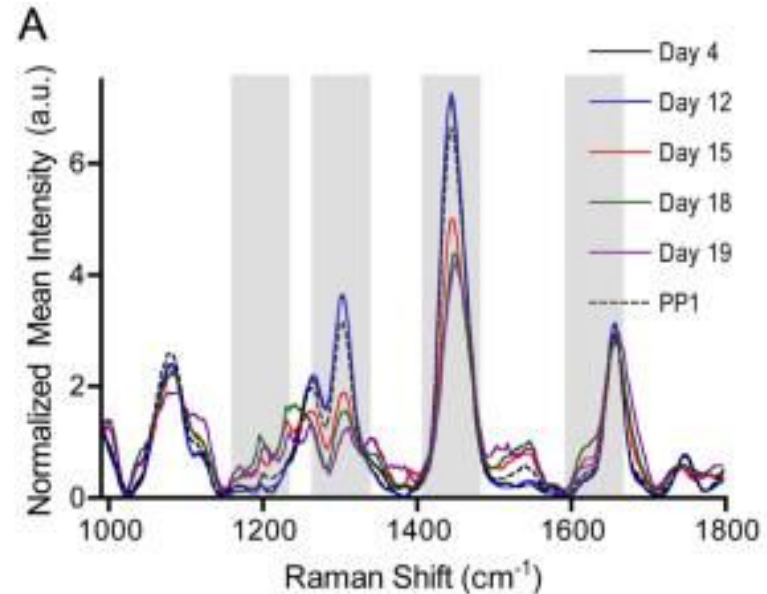
12/06/23

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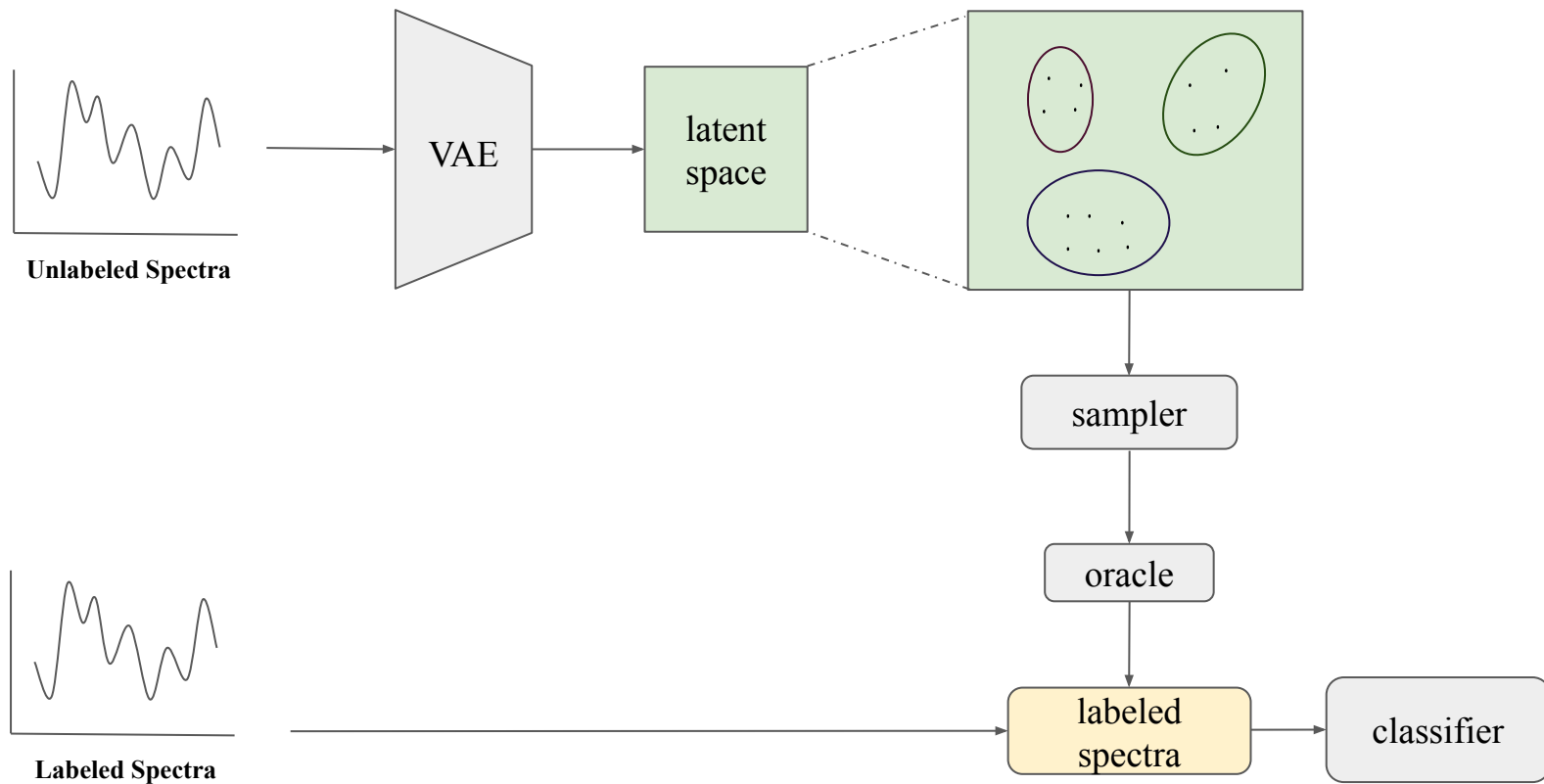
Background: Raman Spectroscopy

- Clinical use: noninvasive tissue examination
 - wavelength vs intensity
- “Tissue” types
 - esophagus
 - human skin
 - chicken breast
 - turkey breast
 - gelatin



O'Brien CM et al. Raman spectroscopy provides a noninvasive approach for determining biochemical composition of the pregnant cervix in vivo. *Acta Paediatr.* (2014).

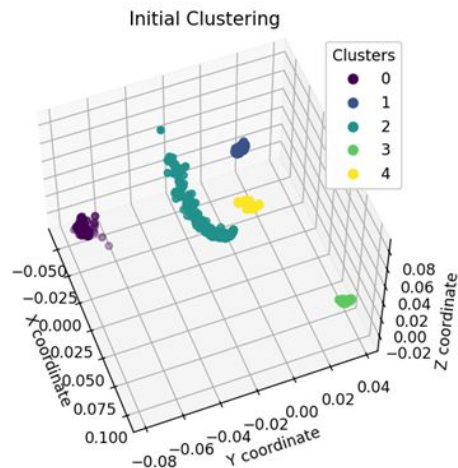
Workflow



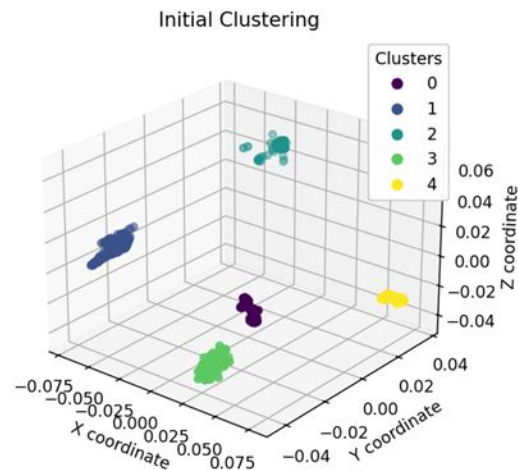
VAE

- Consists of 4 convolutional layers, each with kernel size 3
- Purpose is to compress the data down to easy to plot and cluster data points
 - Compresses down to 3d points
- Reconstruction error will be poor, but we are only compressing
- The VAE will also preserve the important aspects of the spectra

VAE

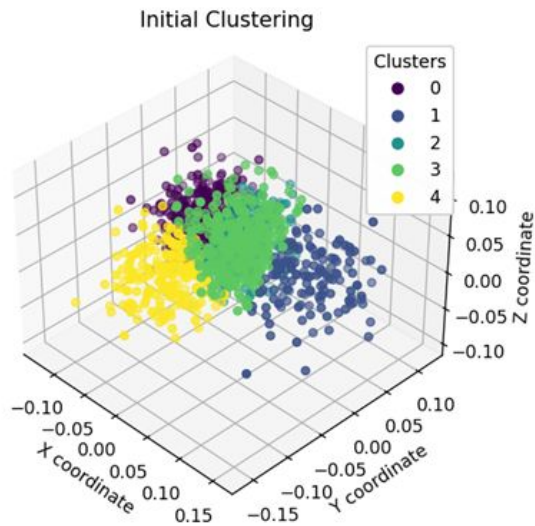


Latent space from VAE using
clean, processed data



Latent space from VAE using
intensity-noised data

VAE (2)



Latent space-noised data

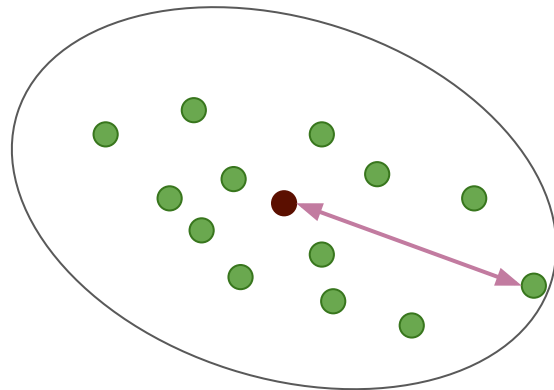
AL & SVM Testing

- Least Confidence Active Learning

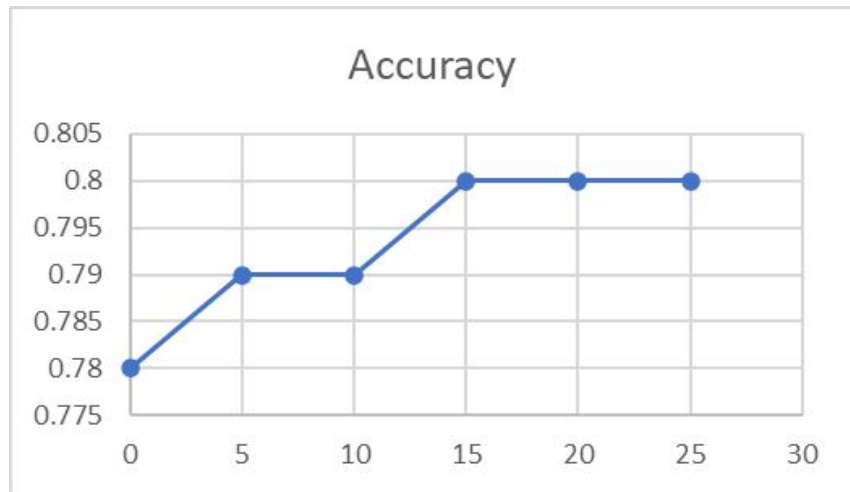
- select n informative samples from each cluster
 - based on greatest distance to cluster center
 - $n = 0, 5, 10, 15, 20, 25$

- SVM

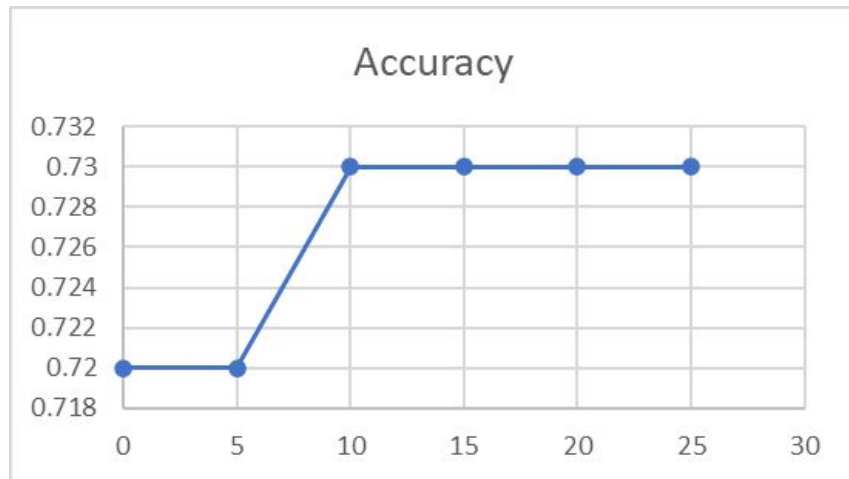
- combine random labeled subset of original data with newly labeled samples
 - 80% train - 20% test split



Discussion

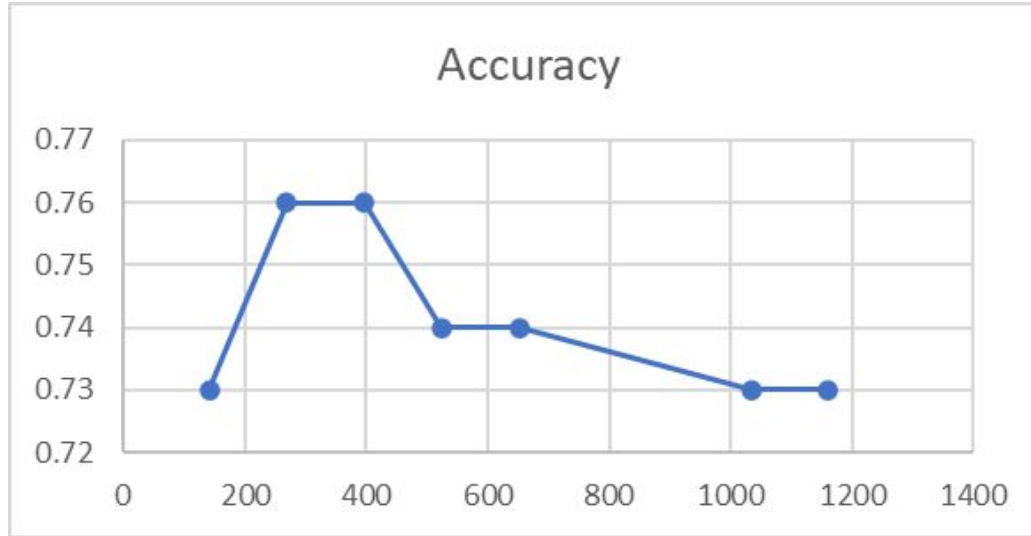


Accuracy vs AL samples on a data set of 269



Accuracy vs AL samples on a data set of 142

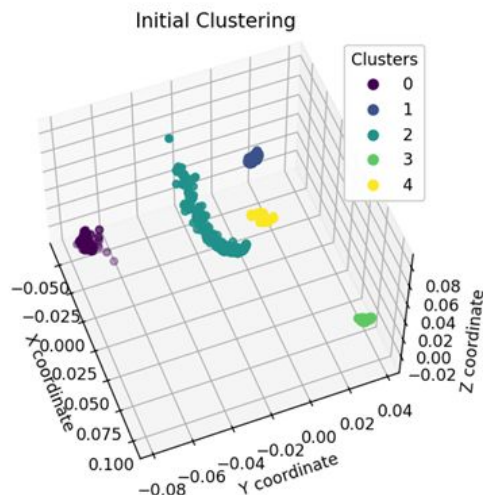
Discussion



Accuracy vs train samples with
15 AL samples

Conclusion

- The active learning helps a little
- The VAE does most of the work
- Medical experts can simply run the VAE and a classification model to provide quality results, AL can be ignored
 - Can also skip the denoising step



Thanks! Questions?